

How ONLINE UPS Can Save You Money



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A logistics enterprise recently switched to an online UPS system after experiencing regular equipment failures. The company had upped its production, which significantly increased its power consumption. While the company did use line-interactive system for power backup, constant voltage fluctuations and overload hampered operations, resulting in unnecessary delays. Understanding the limitation of line-interactive systems to deal with frequent voltage fluctuations and higher loads, the enterprise switched to online UPS system. Since then, it has been a smooth sail for the company with very few hiccups.

The underlying technology

UPS systems can be divided into three categories—online UPS, offline UPS and line-interactive UPS. While offline UPS are mostly used in households to offset power cuts, line-interactive UPS are used in offices for power backup.

There is technically no hard-and-fast limit of the online UPS power capacity. However, generally, online units have capacities between 1kVA and 500kVA due to high-power requirements of industrial equipment and critical devices. In comparison, line-interactive systems are rated below 1kVA, more commonly within 500VA range. Line-interactive systems can also be scaled to support high-power devices, but the increase in price and size and, more importantly, the lack of effective current-correction provision (something we will discuss later)

keep the motive of large-scale backup unattended.

As line-interactive UPS systems cost much less, these are sold more—even at enterprise level. However, large-size companies using expensive and sensitive

devices require higher and cleaner power backup, which makes line-interactive UPS unsuitable for them. Industry-level devices usually require at least 5kVA power capacity, with general requirements ranging above 25kVA, which can go much beyond even 100kVA.

Thus online UPS systems are suggested for large enterprises. These systems can again be divided into two categories—online double-conversion and online delta-conversion.

Online double-conversion UPS systems have two paths for power flow: inverter and bypass. The main power path of this topology is inverter, which means backup power is provided to the inverter—wasting no time in switching in the event of a power cut. Bypass route is used in case the inverter fails to switch on for some reason. Waiting time for switching to the bypass power path is negligible. There is an internal two-step current conversion (AC to DC, followed by DC to smooth AC) that helps in the rectification and smoothing of the inconsistent input voltages before powering the connected device—saving the device from power surges, sags, brownouts or any other kind of power fluctuation. This is the greatest benefit of the online UPS topology for consumers. However, this is also associated with extra energy dissipation, and therefore reduced system efficiency.

Online delta-conversion UPS systems have a comparatively new set of architecture that overcomes the shortcomings of double-conversion mode. While bypass remains the secondary power flow path, just as in double-conversion systems, the delta conversion setup has an additional delta transformer and a delta converter (replacing the battery charger) inserted in the power flow path. Hence, this topology makes direct load connectivity and simultaneous load isolation (in case of any harmful power flow) possible. It also avoids unnecessary power dissipation, increasing the system efficiency. Dur-



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